

The Linguistic Approach: Language and Cognitive Science

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Outline

- Background
- Uniqueness of human language
- Characteristics of human language
- Rule of language
- Role of linguistic in cognitive science

Introduction

- Human language is characterized by a great deal of complexity
 - Challenge to learn a foreign language.... rule complexity
- Acquiring and using it --- possible because complex mechanism in our brain exists to support this activity.
- We are not born with a particular language.
- What is the relation of our language capacity to thinking?

The Uniqueness of Human

- Humans are the only species to engage in language communication
 - Bodily displays other species ... the complex dance of honeybees
 - Vocalization Songs of birds
- Language affords us as much freedom in what we express by means of it as our conscience allows.
- We are able to spare what we express from the requirements of the moments
 - I am on my way to work.... Factual
 - What if I were to hang out in the mall instead..... Speculation
 - I am really sorry I did not come in yesterday, I wasn't feeling well.. 'lies'
- We use language for the large and varied set of intellectual, aesthetic, emotional, and social purposes

Language Flexibility

- Human's language ability are far greater than other species
- Speakers must construct their utterance to take account of the listener's need....
audience design based on listener's general character
- The flexibility we display in conversation is also found in reading

“According to a research at Cambridge University it doesn't matter in what order the letters in a word are. The only important thing is that the first and last letter be at the right place.”

Readers also have the ability to draw inferences to make sense of what they are reading..... based on their knowledge of the world.

The Nature of Language

- Language has five characteristics:
 - **Communicative:** Language allows for communication between individuals. Communication refers to the production, transmission, and comprehension of information.
 - **Arbitrary:** A language consists of a set of symbolic elements... are sounds, pictures, or words.
 - The sound of the word “house” in English AND Spanish.... Arbitrary
 - **Structured:** The ordering of the symbols in a language is not arbitrary, but is governed by a set of rules.
 - **Generative:** The symbolic elements of a language can be combined to create a very large number of meanings.
 - **Dynamic:** Languages are constantly changing as new words are added and grammatical rules altered.
 - *How many words do we have in Amharic Dictionary?*
 - *How many words does the average person know?*

Characteristics of Human Language

- Our language is not made up simply of a set of expressions, rather it is constructed of a fairly small set of sounds, known as **phonemes** nnn, eee, a **phoneme is the smallest unit of sound**
 - String together one after the other to form meaningful bits and pieces: **morphemes and words**.
 - **Morphemes** are the smallest units of spoken language that have meaning.
- Except for a small set of words exhibiting **onomatopoeia** (meow, cuckoo), **the strings of sounds making up our words and parts of words bear no necessary or logical relation to their meaning**.
- **The sounds of words are not logically or necessarily tied to the meaning** ... one of the reasons for flexibility of human language.

Languages are far more similar than we generally perceive.

Characteristics of Human Language

- Meanings
 - Sound sequence may be associated arbitrarily with meanings
 - To express meaning linguistically, organize the sound sequences of a language in to word, **we must arrange them, again in sequence**, into sentences.
- Sentences tend to express complete thoughts
- Although the number of words in a given language is **finite**
 - Yet the number of **sentences that can be constructed out of these words is infinite**.
- The interesting property of human language embodied in sentences is called **recursion**. ... characteristics of all human languages.

The Rule of Language

- Combinations of words, also in a linear order, making sentences
- This is not enough to constitute language
- How do we know the order in which to put them in a meaningful way?
- How do we know that one sounds Amharic and another is not ?

The Rule of Language

- How do we know that one ordering of words is a real sentence and another, using the very same word, is not.
- It cannot be that we have learned each instances individually, sentences are infinite.

The Rule of Language

- The answer is that

“there is finite set of rules that we have learned, rules that enable us to put together the sounds, words, and sentences of our language and to recognize when they are not being followed.”

- If we know the rules, how did we learn them?

The Rule of Language

- Grammar
 - Who tell you how you are supposed and not supposed to say what you wish to express
 - Such rules are called **prescriptive rules**
 - “say ‘yes’, ‘no’, “; a pronoun must agree in gender and number with the noun to which it refers.... Prescriptions for speaking the language in the manner educated people consider correct.
 - **Prescriptive rules may need to be thought**
 - Linguists focus on when they are concerned with a **speaker’s implicit knowledge of linguistic system** describe these rules.... **do not prescribe them**. ... descriptive linguistics ... **not thought**

Why are human languages skill dramatically better than those of any other species?

The Rule of Language

- Our linguistic competence contains the rules pertaining to every component of the language
 - Phonology ... the rules pertaining to the sound system
 - Morphology The rules governing word structure
 - Syntax The rules governing the structure of sentences... arrangement of words in a sentence
 - Semantics The rules concerning meaning
- Collectively, these rules are known as the grammar of the language.
- The rules that underlie our linguistic competence can be deduced or inferred by studying the patterns observable in a language.

Noam Chomsky proposed in the '50s and '60s that humans have an innate capability for learning language that is based upon a pre-wired universal grammar

Role of Linguistics in Cognitive Science

Noam Chomsky claimed that all languages contain similar structures and rules, noting that across the world children acquire language seemingly effortlessly, suggesting that we're born with the basics already present in our brains.

Language has had a central place in the emerging field of cognitive science

- Linguists search for the underlying commonalities among languages
 - Contribute to our knowledge of the types of rules and principles underlying languages.
- Language is of great interest to those in the field of cognitive psychology --- the representation of knowledge in the brain
- Some of the areas in cognitive psychology in which language plays a major role will be discussed in the following.

Role of Linguistics in Cognitive Science

- Language Acquisition
 - A human being is not born with an ability to speak his or her native language fluently...this ability develops over time.
 - Human beings pass through a series of stages, each one marked by the acquisition of new linguistic skills.
 - One branch of linguistic concerned with issues of first-language acquisition is developmental psycholinguistics
 - The approach taken by today's developmental psycholinguists had its origins in the late 1960s and early 1970s
 - Use longitudinal study... carried out for a long period of time

The formal mathematical tradition in linguistics can be argued to have been established by Chomsky

Role of Linguistics in Cognitive Science

- Careful and detailed observation and recording of the linguistic behavior of numbers of children over time..... progress from one level of development to the next.
- Prenatal Development
 - Language learning appears to begin before birth.
 - During the last trimester of prenatal development the fetus can hear its mother's voice as shown by changes in fetal heart rate and motor activity when the mother is speaking, and this affect the infant's preferences for language after birth in a number of ways. (Karmiloff & Karmiloff-Smith, 2001)

Role of Linguistics in Cognitive Science

- Crying

- Babies cry as soon as they are born, a reflexive behavior, not intentional communication from the infant
- The process of communication begins when babies begin to learn that crying can act as a signal that brings relief from hunger, discomfort, and loneliness because it motivates adults to do what it takes to make it stop.

Role of Linguistics in Cognitive Science

- Early in the first year: called cooing stage
 - Babies begin to utter sounds, same in all babies, distinguishable from the crying sound and other vocalizations that occur from birth and indicate various kinds of feelings and needs
 - Not language specific
 - Beginning to use their vocal apparatus... tongues, their lips, their vocal cords... develop into the motor gestures that eventually produce speech

Role of Linguistics in Cognitive Science

- Babbling stage
 - Smaller set of sounds is used..... a good deal of the time into repeated consonant- vowel pairs: baba, mama....
 - The intonational patterns of language also begin to develop in this stage so that though few or none of the utterances are real words
- From utterances.... two word utterances ... then longer strings of words are produced
- The complexity of these utterances develops over time until the basic structure of the language has been acquired.
- Characterizing the grammar that underlies these stages is more difficult

Between 8 and 18 months of age, a major developmental milestone occurs when infants begin to use words to speak.

Role of Linguistics in Cognitive Science

- The social aspect of first-language acquisition
 - Babies learn the language spoken around them
 - We don't give babies formal lessons in how to do it
 - Language is learned through imitation
 - We reinforce the development of a child's language in many ways.
 - Research has shown that the more that mothers respond to their babies' vocalization, the sooner their babies develop language.

Role of Linguistics in Cognitive Science

- How adults foster language development?
 - Adults play in fostering young children's language development by talking to them, even when it is clear that the babies do not understand
 - Adults act as if they do understand and carry on conversations, taking turns with whatever the baby responds

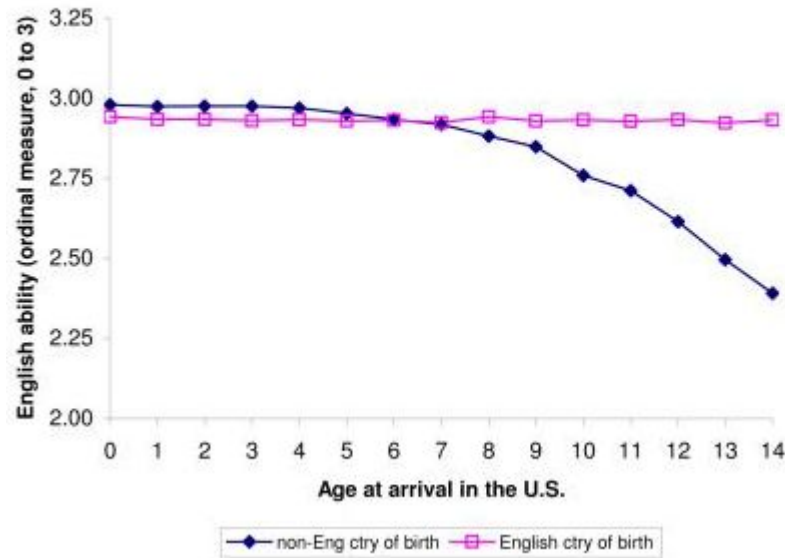
https://www.sagepub.com/sites/default/files/upm-binaries/57593_Chapter_9_Levine_Sample.pdf

Is there a acritical period for language learning?

Role of Linguistics in Cognitive Science

- Second-Language Acquisition
 - The acquisition of subsequent language is also a major area of research
- Involves the learnings of a language or languages after one has acquired a first language

Panel A. Regression-Adjusted Means



- English-Speaking Ability by Age at Arrival

Role of Linguistics in Cognitive Science

- Bilingualism

- Learning to speak a language is a complex cognitive task, so learning to speak two different language is even more cognitively complex.
- The fact that some children acquire more than one first language raises some interesting questions.
 - Do the languages that are being learning simultaneously affect each other, if so, how?
 - Where and how are they stored in the brain?
 - Are patterns of cerebral organization different in bilinguals from those in monolinguals, if so how do they differ?

Role of Linguistics in Cognitive Science

- Bilingualism

- Studies contributed much to our understanding of how the brain handles how language in one speaker.
 - When a bilingual speaker hears one of his or her languages, both languages are activated rather than just the one being heard at the moment
 - The activation of both languages occurs on just when a bilingual individual is hearing one of the language he or she speaks but in fact both language are also activated when the person is speaking one of them
 - Those who have acquired both languages from the very beginning will feel equally at home in both, will not make random language errors, and will be able to switch easily from one to the other

Role of Linguistics in Cognitive Science

- On the other hand, who sometime later have mastered another language and speak both fluently, will feel the first learned to be their dominate language
- In bilinguals both languages are activated in parallel – that is, both are activated during hearing and producing speech – the question arise of how speakers choose the appropriate language in which to respond in a given situation.
- Language selection and control is an aspect of processes associated with the control of action in general
 - This issue is still very much under study

Bilingual advantage is observable not only in children but in your and aging adults as well.

- Studies indicate that adult bilingual performed more efficiently than their monolingual counterparts

Role of Linguistics in Cognitive Science

- Bilingualism

- What language do bilinguals think in or is it both?

<https://www.quora.com/What-language-do-bilinguals-think-in-or-is-it-both> ... Quora

Role of Linguistics in Cognitive Science

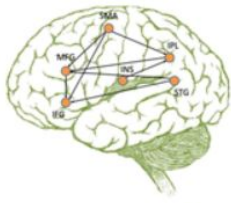
“Actually, *nobody thinks in a certain language*. What happens in your brain is that you think in a stream of concepts that you usually impose a string of words on from a certain language.” Fedor Steeman

Computational Modeling of Bilingual Language Learning

- Computational Modeling of Bilingual Language Learning
 - Any computational account of bilingual language learning needs to be guided by theoretical considerations of how learning gives rise to the bilingual brain and to bilingual cognition and behavior.
 - Bilingual language learning is a complex process that involves multiple domain-specific and domain-general systems.
 - **Connectionist model**: A connectionist model, inspired by neural networks in the human brain
 - In the last three decades-connectionism has become one of the **most influential theoretical frameworks and analytic approaches for understanding human language learning as well as cognitive and linguistic behaviors in general**
 - Example: bilingual interactive activation (BIA) model

Computational Modeling of Bilingual Language Learning

- Computational Modeling of Bilingual Language Learning
 - Pluralist learning mechanisms: combine various computational algorithms and theories to advance the understanding of bilingual language learning
 - Example: integrating both neural and cognitive models -- incorporate cognitive mechanisms and principles that are abstracted away from the detailed biological structures and their computational processes.
- Deep Neural Networks
- Whole-Brain Networks
- Bayesian Statistical Learning

Term	Definition	Example
Node	Element of a network.	
Edge	The connection between two nodes.	
Effective connectivity	A whole-brain network analysis describing the causal influences that neural units exert over each other.	
Community	A group of nodes that are more densely connected to each other than to the other nodes of a network. In the example, there are three communities detected (marked with different colors).	

Role of Linguistics in Cognitive Science

- Language Acquisition in the Deaf and Hearing Impaired
 - Sign language
 - Widely regarded as a mode of communication interior to spoken language
 - Studies of sign languages are indeed languages in the same sense that spoken language is
- Being exposed to spoken language is not the only means to normal linguistic development
 - We can work to enrich the environment of deaf and hearing-impaired children by including complete access to language, allowing them to develop their cognitive abilities as much as they would have had they been able to hear.

Role of Linguistics in Cognitive Science

- Language Loss :
 - Another important avenue to understand both the nature of language and the function of the brain
 - May occur in:
 - Move from one country to another in which a new language is used for communication
 - The language used significantly less may become less available to the individual
- A loss of individual's native language is referred to as first- language - attrition
- The loss of language acquired subsequent to one's native language is known as second-language-attrition

Role of Linguistics in Cognitive Science

- Language Loss : cause of language loss
 - Language loss occurs in people who have suffered brain damage either through disease, such as Alzheimer's disease or brain tumor, trauma.. Stroke or a blow to the head the condition this people exhibit is know as aphasia, like infant this term means lacking speech.
 - The extent and location of damage to the brain determine the aphasic patient's language deficits.

Role of Linguistics in Cognitive Science

- Language and Thought
 - What is the relation of language to thought?
 - The very notion of thinking is abstract; we cannot see it, though facial expressions may give us a clue that it is occurring.
 - The capacity for abstract thought depends on the capacity for language.
 - An important question in this regard concerns the sort of language in which our thoughts are encoded.

Role of Linguistics in Cognitive Science

- Language and Thought
 - Do we think in ordinary language?
 - Example: How often has it happened to you that you sought a word to express an idea, knowing you had the word somewhere, that it was on the tip of your tongue, but you just couldn't find it?
 - What about the time your thoughts were racing faster than you could put them into words?
 - When asked to define a work, "I know what it means; I just don't know how to explain it"

Role of Linguistics in Cognitive Science

- Thoughts that are ultimately expressed in language exist somehow in advance of their being so expressed.
- Fodor published a book “ The Language of Thought” .. Fodor hypothesized that mental representation has a linguistic structure, that it is a symbolic system realized physically in the brain
 - Important contribution to the subject

Role of Linguistics in Cognitive Science

- Computational Linguistics

- The role of computation linguistics, like that of cognitive science, is **interdisciplinary**, drawing on the work of researchers in a number of fields

- Research directions

- **Theoretical models**: approach to understand the underline rule of any language
 - **Practical use** of modeling human language: creating machines that can work with humans using language to communicate
 - **Statistical learning** is the capacity of humans, babies included, to observe the statistical regularities inherent in the information they receive from the environment to their brains.

Further reading: Saffran JR. Statistical language learning in infancy. Child Dev Perspect. 2020 Jan 19;14(1):49-54. doi: 10.1111/cdep.12355. PMID: 33912228; PMCID: PMC8078161.

Role of Linguistics in Cognitive Science

- Computational Modeling of Language
 - An important line of research in the last couple of decades
 - Was developed by researchers in AI
 - A good example: it was demonstrated that a particular linguistic phenomenon, such as the learning of the past tense of verbs, could indeed occur in such a model
- There are major difference between any computational model and the functioning of the human brain
 - Metaphorically in referring to aspects of brain functions:
 - Hardware consists in the physical component of the brain
 - Software; the neural process that our brain engage in
 - Computer programs are written run serially, as opposed to the parallel processing that takes palace in our brain

Artificial Intelligence and Linguistics: Natural Language Processing

- Four stages of natural language understanding
 - **Speech recognition**:- the acoustic speech signal is analyzed to determine the sequence of spoken words.
 - **Syntactic analysis**:- the word sequence is analyzed via the use of knowledge of the language's grammar.
 - **Semantic analysis stage**:- derive a partial representation of the meaning of a sentence using the sentence structure and the meanings of the words
 - **Pragmatic analysis**:- the final stage, produces a complete meaning for the sentence via the application of contextual information.
 - This information includes data that has to do with the time and location of the utterance, who was saying it, and to whom it was said.

Artificial Intelligence and Linguistics: Natural Language Processing

“Human languages are notoriously complex, and linguists have long thought it would be impossible to teach a machine how to analyze speech sounds and word structures in the way human investigators do.”

MIT News: <https://bcs.mit.edu/news/ai-can-learn-patterns-human-language>

Computational linguistics traditionally uses computational models to address questions in linguistics and borders the field of natural language processing.

Language Models

- Language Models (LMs) has been rooted in the quest to understand and replicate human language
 - Statistical Language Models: capture the statistical properties of language, such as word frequencies and co-occurrences, to predict the likelihood of a given sequence of words based on the Markov assumption.
 - Limited by the exponential number of transition probabilities to be estimated and the Markov assumption
 - Not always hold true in the complexity of natural languages.
 - Neural Language Models: the development of language models that utilised neural architectures to capture language's complex patterns and dependencies.
 - Recurrent neural networks (RNNs) and long short-term memory (LSTM) networks, could capture long-range dependencies and contextual information
 - Later, word2vec introduced -- a shallow, two-layer neural network trained to reconstruct the linguistic contexts of words
 - A significant leap --- representing a shift from word sequencing to learning representation.

Language Models

- Language Models (LMs) has been rooted in the quest to understand and replicate human language
 - Pre-trained language models: The development of pre-trained language models marked a significant milestone in the evolution of language models.
 - Trained on large data corpora in an unsupervised or self-supervised manner before being fine-tuned on specific tasks.

“The idea is to pre-train a model on a diverse data set and then transfer its knowledge to a narrower task by fine-tuning it on a smaller, task-specific dataset.”

- BERT (Bidirectional Encoder Representations from Transformers): a transformer-based model pre-trained on a large corpus of text and then fine-tuned it on specific tasks.
 - A significant advancement in natural language processing
 - Demonstrated the potential of pre-trained language models to achieve state-of-the-art performance on a wide range of tasks.
 - Inspired a significant number of models, such as GPT-2, GPT-3

Language Models

- Large Language Models
 - Redefined the capabilities of language processing systems
 - Built on the **transformer architecture**— designed to capture long-range dependencies and contextual information in language
 - Exhibit remarkable performance **text generation, question answering, translation, and summarization**, often rivaling human-like comprehension.
 - Beyond their core functions, showing proficiency in tasks like **commonsense reasoning, code generation, and arithmetic**.

ChatGPT: an Advanced Language Model

- Developed by [OpenAI](#)
- Is an AI language model that is trained on large amounts of text data to learn patterns and generate responses
- ~~GTP~~-[Generative Pre-trained Transformer](#)
 - Is a neural network-based model that uses self-attention mechanisms to process text data
- Designed to generate human-like responses to [text-based inputs](#)

ChatGPT: an Advanced Language Model

- Used in a **variety of applications:**

- Language translation,
- Text summarization, and
- Content creation

- **Limitations**

- The potential for bias in the text data it is trained on, which can lead to biased responses

- Focus on pre-training, fine-tuning, generation

“One of the key features of ChatGPT’s pre-training process is the use of **unsupervised learning**. This means that the model is not explicitly taught how to respond to specific inputs or outputs but instead learns through exposure to large amounts of text data.”

The search for **computational models inspired by infant-like learning**– BabyLM-- gives machine learning researchers the task of training language models from scratch on amounts of linguistic data similar to those available to a 13-year-old child: around 100 million words, rather than the estimated 300 billion words ingested by ChatGPT.

- Debate on whether LLMs are truly capable of understanding language or merely mimic it--- **LLMs currently have little to do with linguistics and human cognition**, and there is a chance that in the future they will diverge even more

Advanced Language Models

COMPARATIVE ANALYSIS OF FOUR MAIN LARGE LANGUAGE MODELS (LLMs)

CRITERIA	 Google Gemini	 OpenAI GPT-4	 Anthropic Claude 3	 Meta Llama 3
KEY STRENGTHS	 NATIVE MULTIMODAL (TEXT, IMAGE, VIDEO, AUDIO)	 STRONG LOGIC & CODING ABILITIES	 FOCUSED ON SAFETY & ETHICAL AI	 LEADING OPEN WEIGHTS MODEL
PRIMARY USE CASES	 MASSIVE CONTEXT WINDOW (1M+ TOKENS)	 REAL-TIME OMNI-MODAL INTERACTION	 HUMAN-LIKE WRITING & NUANCE	 RESEARCH-ORIENTED APPLICATIONS
UNIQUE FEATURES	 ADVANCED REASONING	 WIDE ACCESSIBILITY (ChatGPT)	 HIGH PERFORMANCE ON LONG DOCUMENTS	 LOCAL & EFFICIENT DEPLOYATIONS
MODEL VERSIONS	 GEMINI GOOGLE - GEMINI	 GPT-4 GPT-4	 ANTH CLAUDE 3	 OPEN LLAMA 3

Language Models

- Small Language Models:
 - Rapidly emerging subset of artificial intelligence designed
 - Operate with a fraction of the parameters used by LLMs
 - This reduction in size allows them to function in resource constrained environments such as edge devices, mobile platforms, and offline scenarios
 - Like their larger counterparts, leverage a transformer architecture.

Language Models

- **Small Language Models:**

- To reduce model size while retaining functionality, model compression techniques are applied.
 - **Pruning:** Eliminating redundant parameters from neural networks to simplify computations while preserving core performance.
 - **Quantization:** Representing model weights and activations in lower precision (e.g., 8-bit integers) to improve speed and reduce memory usage.
 - **Low-Rank Factorization:** Decomposing weight matrices into simpler approximations to lower computational demands.
 - **Knowledge Distillation:** Transferring knowledge from larger “teacher models” to smaller “student models”, enabling compact versions to retain critical features.

Models: DistilBERT, Google Gemma, Minstral and

others

Language Models

“LLMs are not without challenges, including ethical concerns, environmental impact, and the potential for bias and hallucination in generated text: Matarazzo and Torlone (2025)”